Ariyan Zarei

Senior Computer Vision Engineer | 3D Reconstruction | Human-Centric Vision | Neural Networks

Burlingame, CA arianzarei73@gmail.com linkedin.com/in/ariyanzarei

github.com/ariyanzri ORCID: 0000-0002-3670-2472

Professional Summary

Senior computer vision engineer and research scientist with a Ph.D. in Computer Science and industry experience at HOVER and Meta Reality Labs. Focused on 3D reconstruction, human-centric vision, motion tracking, geometry registration, neural network training, and scalable data pipelines across NeRFs, Gaussian splats, meshes, point clouds, and voxels. Strong record of taking ambiguous research problems from formulation through implementation, large-scale processing, evaluation, publication, and cross-functional delivery.

Core Strengths

- Human motion, body, and hand tracking
- 3D computer vision and geometry processing
- Large-scale ML and CV data pipelines
- Deep learning for images, 3D representations, and spatio-temporal data
- Photogrammetry, registration, and reconstruction
- End-to-end research and software delivery

Professional Experience

Senior Computer Vision Engineer August 2026 – Present
HOVER *San Francisco, CA*

- Develop 3D reconstruction systems that recover accurate building models from multimodal imagery.
- Work across image understanding, geometry, and reconstruction pipelines to transform heterogeneous visual inputs into structured 3D assets.

Research Scientist May 2023 – August 2026
Meta Reality Labs *Burlingame, CA*

- Led development of a large-scale full-body human motion tracking and geometry registration pipeline for the Codec Avatars program, supporting billion-scale internet in-the-wild data processing.
- Designed and built hand tracking pipelines for multi-view studio capture and monocular mobile datasets, connecting data preparation, tracking, registration, and downstream research needs.
- Drove end-to-end execution across research, data, and infrastructure partners, turning open-ended avatar-processing requirements into reliable, scalable pipeline components.

Machine Learning Software Engineer Intern May 2022 – August 2022
Meta *Menlo Park, CA*

- Worked with the Ads Fairness team on modeling strategies for group-level calibration and fairness-sensitive decision making.
- Designed and developed a policy-gradient reinforcement learning controller and an aggregated supervised learning model to address potential miscalibrations across user groups.

Graduate Research Assistant January 2019 – April 2023
University of Arizona *Tucson, AZ*

- Created MegaStitch, a robust and scalable geo-correction and image stitching method for agricultural orthomosaics, published in *IEEE Transactions on Geoscience and Remote Sensing*.
- Developed MegaStitch3D for geo-correcting large 3D point clouds in field phenotyping workflows with the PhytoOracle and CyVerse teams.

- Built PlantSegNet, an end-to-end deep instance segmentation model for 3D point clouds of sorghum leaves, later published in *Computers and Electronics in Agriculture*.
- Designed NowCasting-Nets, convolutional and recurrent neural networks for estimating precipitation patterns from spatio-temporal satellite data.

Additional Experience: IT Intern, NP Photonics (2021–2022); Undergraduate Research Assistant, Shahid Beheshti University (2016–2017).

Education

Ph.D. in Computer Science — August 2018 – May 2023
University of Arizona; Advisor: Prof. Kobus Barnard — Tucson, AZ

M.Sc. in Computer Science — August 2018 – December 2020
University of Arizona — Tucson, AZ

B.S. in Computer Science — August 2013 – April 2018
Shahid Beheshti University — Tehran, Iran

Technical Skills

Machine Learning and CV	PyTorch, PyTorch3D, TensorFlow, Keras, OpenCV, neural network training and evaluation
3D Vision and Geometry	Motion tracking, registration, reconstruction, photogrammetry, NeRFs, Gaussian splats, meshes, point clouds, voxels, neural rendering
Software Engineering	Python, C++, C, C#, Java, Docker, Singularity, Django REST Framework, React Native
Data and Systems	Large-scale data pipelines, Microsoft SQL Server, Windows Server, Active Directory, Office 365
AI-Assisted Engineering	Codex, Claude, Claude CLI, agentic coding workflows for rapid prototyping and project development
Hardware Prototyping	Raspberry Pi, Raspberry Pi Pico, ESP32, MicroPython, embedded circuits

Selected Publications

1. "Large-scale Codec Avatars: The Unreasonable Effectiveness of Large-scale Avatar Pretraining." *arXiv*, 2026.
2. "Head and Body Pose Estimation using Ultrasound Sensing on Smart Glasses." *CHI Extended Abstracts*, 2026.
3. "MegaStitch: Robust Large-Scale Image Stitching." *IEEE Transactions on Geoscience and Remote Sensing*, 2022.
4. "PlantSegNet: 3D point cloud instance segmentation of nearby plant organs with identical semantics." *Computers and Electronics in Agriculture*, 2024.

Additional publications in satellite nowcasting, phenomics, and scalable data pipelines are listed on ORCID.

Awards, Leadership, and Languages

- WorldSkills bronze medal, IT Software Solutions.
- Tech Launch Arizona Award, intelligent gardening application.
- 2022 Galileo Circle Scholarship, University of Arizona.
- Student club leadership, University of Arizona and SBU.

Languages: English, Persian.